

Datorix Backup + Replication

Encrypted, deduplicated database backups to S3-compatible storage and governed CDC replication to ClickHouse — both orchestrated from a single management console with full audit logging.

AES-256 CLIENT-SIDE ENCRYPTION	5 min TO FIRST BACKUP JOB	7-yr AUDIT RETENTION (SOX)	6+ S3-COMPATIBLE TARGETS
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PRODUCT OVERVIEW

Database backups are universally mandated and universally neglected. Unencrypted snapshots on local disk, untested restores, no incremental coverage, and disconnected replication mean most enterprises discover backup failures only during a recovery — the worst possible time.

Datorix Backup + Replication delivers encrypted, deduplicated database backups to any S3-compatible object storage via Kopia, and a governed Change Data Capture (CDC) replication pipeline from any source database to ClickHouse — both orchestrated from a single management console with RBAC enforcement and complete execution audit logging.

FEATURES & BUSINESS BENEFITS

Database Backup

- **PostgreSQL Backup:** pg_dump (custom format) piped through Kopia for AES-256 client-side encryption and deduplication before reaching S3. Parallel pg_restore supported.
- **ClickHouse Backup:** Native BACKUP DATABASE ... TO S3(...) via ClickHouse HTTP interface. No agent required on the ClickHouse host.
- **Truncation Protection:** If pg_dump exits non-zero, the orphaned Kopia snapshot is immediately deleted — no partial dumps retained as valid backups.
- **Backup Verification:** Quarterly DR procedure: Kopia restore into a drill schema, sanity queries, and a full audit report. An untested backup is not a backup.

CDC Replication

- **PostgreSQL to ClickHouse:** Debezium reads committed changes from the PostgreSQL WAL, publishes to Kafka (dam.cdc), and the Replication Runtime auto-provisions ClickHouse targets.
- **Auto-Provisioned Targets:** Replication Runtime creates rt_raw_* (Kafka engine), rt_mv_* (materialized view), and rt_final_* (MergeTree) tables — no manual DDL.
- **Replication Health Monitoring:** Commander reconciles job state on a configurable schedule and surfaces DEGRADED status when replication lag exceeds the threshold.
- **Planned Sources:** MySQL, Oracle, and Microsoft SQL Server CDC via Debezium connectors — provisioning model identical to PostgreSQL.

Data Security & Encryption

- **AES-256 Client-Side Encryption:** Kopia encrypts data before it leaves the host. Credentials are never passed as command-line arguments — no secrets in process lists.
- **S3-Compatible Object Storage:** MinIO (on-premises), Amazon S3, Wasabi, Backblaze B2, Garage, and Ceph RGW. Storage endpoint is swappable without changing backup job definitions.
- **RBAC Enforcement:** SUPER_ADMIN / ADMIN / OPERATOR roles enforced on every Commander REST endpoint. No job can be created, modified, or deleted without appropriate privileges.
- **Immutable Execution Log:** Every backup run logged: status (PENDING → RUNNING → SUCCEEDED/FAILED), duration, bytes processed, snapshot ID, and error excerpt if failed.

Monitoring & Governance

- **Execution Audit Trail:** Every job run logged with full metadata: timestamps, bytes transferred, snapshot ID, status, and error excerpt — all queryable from the console.
- **Health Reconciler:** Commander @Scheduled reconciler polls job state and surfaces DEGRADED when last_run_at becomes stale beyond the configured threshold.
- **Alert Notifications:** Route backup failure and replication lag alerts to Slack, PagerDuty, Microsoft Teams, and email with configurable escalation policies.
- **Report Export:** Export backup execution history and replication status as PDF or CSV for compliance teams and disaster recovery audits.

Retention & Recovery

- **Configurable Retention:** Daily / weekly / monthly retention per job. Defaults: 7 daily, 8 weekly, 12 monthly. Automatic pruning runs after each successful backup.
- **Restore from Snapshot:** Restore any PostgreSQL dump to a target DSN directly from Commander via Kopia restore streamed to pg_restore. Snapshot list queryable from the UI.
- **Job Scheduling:** In-process cron with missed-run catch-up on restart. No Kubernetes CronJob permissions or static manifests required.
- **Point-in-Time Recovery [Roadmap]:** WAL archiving (pg_basebackup + continuous WAL shipping to S3) will enable restore to any second within the retention window.

Enterprise Administration

- **Unified Control Plane:** Commander API manages backup jobs, replication connectors, and storage locations via a single REST API with consistent RBAC enforcement.
- **API Key Management :** Scoped, rotatable API keys for programmatic job creation, status polling, and restore triggers — each key carries its own permission scope.
- **MFA — TOTP :** Time-based one-time passwords (RFC 6238) for all console users. Enforcement configurable per role by the SUPER_ADMIN.
- **SSO / LDAP / Active Directory:** OAuth2/OIDC federation and LDAP bind against Microsoft AD, Okta, and Auth0. Group-based role provisioning on login.
- **High Availability :** Active-passive Commander clustering with automatic failover for zero-downtime backup orchestration.

SYSTEM ARCHITECTURE: THE 5-STEP PIPELINE

01 DUMP pg_dump or native ClickHouse BACKUP captures database state on the configured cron schedule.	02 ENCRYPT Kopia applies AES-256 client-side encryption and deduplication before any data leaves the host.	03 STORE Encrypted snapshots land in S3-compatible object storage: MinIO, Amazon S3, Wasabi, or Backblaze B2.	04 REPLICATE Debezium reads WAL changes, Kafka delivers them, Replication Runtime provisions ClickHouse MergeTree targets.	05 AUDIT Every execution logged: status, duration, bytes, snapshot ID. Full trail queryable from the console.
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TECHNICAL SPECIFICATIONS & INTEGRATION STACK

In addition to snapshot-based backups, Datorix leverages highly efficient Change Data Capture (CDC) streaming pipelines. By parsing underlying database transaction logs without triggers or schema changes, Datorix ensures zero disruption to business-critical tables.

COMPONENT	TECHNOLOGY	SPECIFICATION
Backup Engine	Kopia	AES-256 client-side encryption, deduplication, snapshot management, S3-compatible backends
PostgreSQL Backup	pg_dump	Custom format; parallel restore via pg_restore; truncation protection on non-zero exit code
ClickHouse Backup	Native BACKUP TO S3	No agent on ClickHouse host required; daily and weekly cadence configurable per job
Object Storage	S3-compatible API	MinIO, Amazon S3, Wasabi, Backblaze B2, Garage, Ceph RGW — endpoint swappable per job
CDC Pipeline	Debezium + Kafka	WAL/binlog reader → Kafka → ClickHouse Kafka engine → rt_raw_*, rt_mv_*, rt_final_* MergeTree
Control Plane	Commander API (Java)	Job CRUD, RBAC (SUPER_ADMIN/ADMIN/OPERATOR), @Scheduled reconcile, REST callbacks
Backup Runtime	Go 1.23 / Alpine	Static binary; multi-stage Alpine container; non-root user; postgresql16-client included

FLEXIBLE DEPLOYMENT ARCHITECTURES

Backup Runtime

Standalone Go static binary — no CGO, no host dependencies. Deployed as a multi-stage Alpine container alongside the Commander API. Connects to any PostgreSQL or ClickHouse target over the network. No agent installation on the database host required.

Replication Runtime

Requires a Kafka Connect cluster with Debezium source connector (provisioned automatically via Kafka Connect REST API). ClickHouse receives committed row changes via the Kafka engine into auto-created MergeTree tables. Runs at replicas=1 with Recreate strategy to enforce single-writer consistency.

Corporate Presence

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Next Steps

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